

Rank Correlation Meets Gaussian Degree Normalization in GNNs for Image Classification

Rafael Mendonça Duarte*, Gabriel Maia Brito*, and Lucas Pascotti Valem

Institute of Mathematics and Computer Science (ICMC), University of São Paulo (USP), São Carlos, Brazil

Email: rmduarte@usp.br, gabrielmaiab@usp.br, lucas@icmc.usp.br

*These authors contributed equally to this work.

Abstract—Semi-supervised learning is increasingly important for image classification, where labeled samples are scarce but unlabeled data is abundant. Graph Neural Networks (GNNs) are well suited to this setting, propagating label information from a few labeled instances across a graph that encodes how samples relate to one another. Their application to visual data, however, is challenging due to several factors, with two being the focus of this work. First, images often lack an explicit relational structure, so the graph is typically computed from feature similarity, most commonly through k -nearest neighbor (kNN) strategies. Second, many GNNs normalize message passing using the node degree, which counts a node’s neighbors and, in unweighted similarity graphs, treats all of them as equally relevant, ignoring meaningful differences between them. For image data, graph construction and degree normalization are typically studied as separate design choices rather than redesigned together. While adaptive graph-structure learning and attention-based reweighting have been explored, they are often evaluated primarily on graph-native data (citation and social networks). In this paper, we propose D-GRaNDe, which combines two recent rank-based approaches. It builds the input graph with the Density-Guided Correlation Graph (DGCG) that computes edges from the rank correlation between image similarity rankings. Message passing is then normalized with the Gaussian Rank-based Neighborhood Degree (GRaNDe), which combines a node’s neighbor count with the Gaussian-weighted distances to those neighbors. We evaluate D-GRaNDe against different graph constructions and several centrality-based degree measures across three datasets using only 10% of labeled data. In most cases, D-GRaNDe outperforms both of its individual components and several traditional and recent baselines for semi-supervised image classification. The source code is publicly available at [dgrande.lucasvalem.com](https://github.com/dgrande/lucasvalem.com).

I. INTRODUCTION

While the volume of data produced across virtually every domain continues to grow at an accelerating pace, manual annotation remains a critical bottleneck due to its prohibitive cost [1]. As a result, the number of labeled samples is limited in many real-world applications. In this context, semi-supervised learning has gained increasing attention, as it aims to exploit both the scarce labeled data and the abundant unlabeled data during learning [2].

Among the approaches suitable for this setting, Graph Neural Networks (GNNs) stand out for their ability to exploit relational structure and propagate label information from a small set of labeled instances to many unlabeled ones [3]. The recent rise of foundation models naturally complements these methods. By using the general-purpose representations of foundation models [4] as node features, GNNs can better capture how samples relate to one another and propagate labels efficiently, even under label scarcity. Even alongside newer architectures such as Graph Transformers [5], traditional GNNs remain competitive with the state-of-the-art [6]. However, since their effectiveness still relies on graph topology, effective graph modeling remains a critical factor for success.

This challenge of graph modeling is amplified in image domains, where most datasets lack inherent structural relation-

ships, and the graph is typically built from feature similarity instead. In this setting, most approaches rely on k -nearest neighbor (kNN) or reciprocal kNN strategies [7], and improper construction hinders the propagation of relevant information between nodes, directly impacting effectiveness.

A second, complementary limitation has received far less attention. Many traditional GNN architectures normalize propagation through the node degree, but the standard degree relies solely on connectivity and implicitly assumes uniform importance among neighbors [8]. While alternative notions of node importance exist, they target domains where the graph is readily available [9], and few works revisit the degree for graphs derived from visual data, where edges encode feature similarity and neighbor relevance is far from uniform [9].

Two recent approaches address these challenges independently. The Density-Guided Correlation Graph (DGCG) [10] builds the input graph from rank correlation measures between image similarity rankings, using a density-guided strategy to automatically select the correlation threshold that controls sparsity. The Gaussian Rank-based Neighborhood Degree (GRaNDe) [11] replaces the standard degree centrality with a measure that combines neighborhood ranking and Gaussian distance weighting, emphasizing closer, more relevant neighbors while attenuating distant ones. Although both leverage *contextual similarity*, which measures image similarity from the agreement of their neighborhood relations rather than from raw pairwise feature distance, they act on different components of the GNN workflow and have not been investigated jointly.

In this work, we introduce **D-GRaNDe**, a unified framework that employs DGCG to construct the input correlation graph for the GNN, and uses GRaNDe as the degree function to guide message-passing propagation. Our hypothesis is that by combining these approaches, we can further improve the effectiveness of semi-supervised classification.

The main contributions of this work are as follows:

- We propose D-GRaNDe, a unified approach that combines DGCG [10] with GRaNDe [11], integrating contextual similarity into both graph construction and degree normalization.
- An important finding is that rank correlation graphs and degree normalization are complementary: to the best of our knowledge, their interaction has not been previously investigated, and our ablation shows that the gains from GRaNDe depend on graph topology, with the integration generally outperforming either technique alone.
- We conduct a comprehensive evaluation on multiple image datasets, features, and GNN models in a semi-supervised setting, showing that D-GRaNDe outperforms traditional and recent baselines.

II. PROPOSED APPROACH

The main steps of D-GRaNDe are summarized in Figure 1. Step A performs the feature extraction and the computation

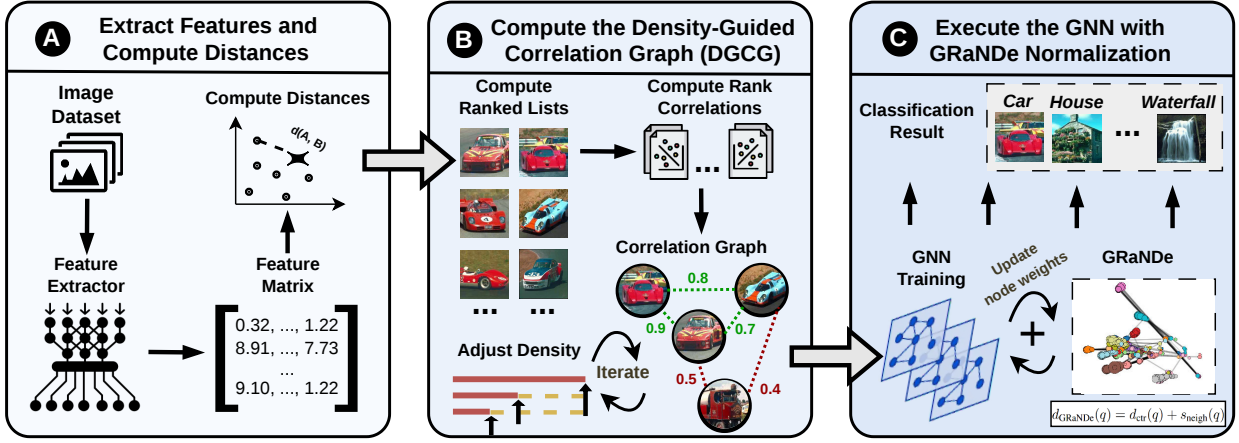


Fig. 1. Overview of the proposed D-GRaNDe approach and its main steps for semi-supervised image classification.

of distances (Section II-A); step B describes how the graph is constructed (Section II-B); and, finally, step C trains the GNN using GRaNDe as the degree normalization (Section II-C).

A. Extract Features and Compute Distances

In this section, we describe how the image dataset is transformed into ranked lists, which we formalize as follows. Let $\mathcal{C} = \{x_1, x_2, \dots, x_n\}$ be a dataset of n images. Our formulation of ranked lists is motivated by image retrieval tasks, where the goal is to retrieve, given a query image x_q , the elements of $\mathcal{C}$ most visually similar to it. Comparison is performed over numerical representations rather than the raw images: a feature descriptor is a function $f: \mathcal{C} \rightarrow \mathbb{R}^d$ that encodes each image as a d -dimensional vector, so that $\mathbf{x}_i = f(x_i) = [x_{i1}, x_{i2}, \dots, x_{id}]$, with $x_{ij} \in \mathbb{R}$, and applying f to the whole collection generates the feature set $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\} \subset \mathbb{R}^d$.

Dissimilarity between two feature vectors is measured by a distance function $\rho: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$, where a smaller value of $\rho(\mathbf{x}_i, \mathbf{x}_j)$ indicates greater similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$. In this work, ρ is the Euclidean distance, following the protocol originally adopted by GRaNDe [11], which also computes its Gaussian neighborhood penalty over Euclidean distances in the learned latent space. The k nearest neighbors of a query $\mathbf{x}_q$ form the set $\mathcal{N}(\mathbf{x}_q, k) \subseteq \mathcal{X}$, i.e., the k vectors closest to $\mathbf{x}_q$ under ρ . For simplicity, we denote this neighborhood set as $\mathcal{N}(q, k)$. Sorting $\mathcal{X}$ by increasing distance to $\mathbf{x}_q$ produces the ranked list τ_q , a permutation of $\mathcal{X}$ in which $\tau_q(\mathbf{x}_i)$ denotes the position (rank) of $\mathbf{x}_i$, so that a lower rank indicates greater similarity and $\tau_q(\mathbf{x}_i) < \tau_q(\mathbf{x}_j)$ implies $\rho(\mathbf{x}_q, \mathbf{x}_i) \leq \rho(\mathbf{x}_q, \mathbf{x}_j)$. We repeat this procedure, taking each feature vector $\mathbf{x}_i \in \mathcal{X}$ as the query, which produces a ranked list τ_i for every image in the collection. The ranked lists of all images form the set $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_n\}$.

B. Compute the Density-Guided Correlation Graph (DGCG)

Given the ranked lists obtained previously, we can now build the graph $G = (V, E)$ that serves as input to the GNN. We define the nodes by treating each image as a node, that is, $V = \mathcal{C}$. The way edges are modeled, however, varies across the literature. For instance, the kNN graph connects every node to its k nearest neighbors, i.e., $E_{knn} = \{(i, j) \mid j \in \mathcal{N}(i, k)\}$. A well-known variant is the reciprocal kNN, which keeps an edge only when both nodes appear in each other's neighborhood, i.e., $E_{rec} = \{(i, j) \mid j \in \mathcal{N}(i, k) \wedge i \in \mathcal{N}(j, k)\}$.

In this work, we adopt the Density-Guided Correlation Graph (DGCG) [10], which measures the correlation between

ranked lists to define a new similarity and connects only the pairs whose correlation exceeds a threshold θ , i.e.,

$$E_{DGCG} = \{(i, j) \mid j \in \mathcal{N}(i, L) \wedge \lambda(\tau_i, \tau_j, k) > \theta\}.$$

In other words, to construct the graph, for each image x_i we first identify its L nearest neighbors $\mathcal{N}(i, L)$. Then, for every neighbor $x_j \in \mathcal{N}(i, L)$, we compute the rank correlation $\lambda(\tau_i, \tau_j, k)$ between their ranked lists up to a neighborhood depth k . Whenever $\lambda(\tau_i, \tau_j, k)$ exceeds the threshold θ , the directed edge (i, j) is added to E_{DGCG} .

However, building the graph in DGCG involves two important aspects, clarified next: (i) how the rank correlation is computed and (ii) how the threshold is defined.

1) *Rank Correlation Measures*: Different correlation measures can be used to compute the similarity between images. Following DGCG [10], we considered both Rank-Biased Overlap (RBO) [12] and JaccardMax.

• *Rank-Biased Overlap (RBO)*: This measure considers the overlap between top- k lists at increasing depths, weighting each overlap by probabilities defined at every depth. It is formally defined as:

$$\lambda_{RBO}(\tau_i, \tau_j, k) = (1 - \mu) \sum_{d=1}^k \mu^{d-1} \times \frac{|\mathcal{N}(i, d) \cap \mathcal{N}(j, d)|}{d},$$

where $\mathcal{N}(i, k)$ denotes the top- k neighborhood of image x_i and μ is a constant. In this work, we set $\mu = 0.9$.

• *JaccardMax*: It captures the strongest similarity indication across the different depths of the ranked lists. It is defined as the maximum Jaccard similarity computed over all depths $k_d \in [1, k]$:

$$\lambda_{JacMax}(\tau_i, \tau_j, k) = \max_{k_d \in [1, k]} \left\{ \frac{|\mathcal{N}(i, k_d) \cap \mathcal{N}(j, k_d)|}{|\mathcal{N}(i, k_d) \cup \mathcal{N}(j, k_d)|} \right\}.$$

2) *Selecting the Threshold via Graph Density*: Since manually setting the threshold θ is difficult, DGCG assumes that sparser graphs benefit GNNs and controls sparsity indirectly through the graph density $\eta = |E|/(n \times L)$. To fit this density within a target interval, it performs a bisection search over $\theta \in [0, 1]$. We adopt the target interval proposed in the original DGCG paper, $[0.03, 0.04]$, with a maximum of 5 iterations. At each iteration, the correlation graph is rebuilt by connecting nodes whose rank correlation exceeds the current threshold, with θ lowered if the density falls below this interval or raised if it exceeds it. The search stops once $\eta \in [0.03, 0.04]$ or after at most 5 iterations, selecting the threshold automatically.

C. Execute the GNN with GRaNDe Normalization

Graph Neural Networks (GNNs) perform semi-supervised learning by propagating node information across the graph topology. We feed both the feature set $\mathcal{X}$ from step A and the graph from step B to the GNN for training and inference, using two models: (i) **SGC** (Simple Graph Convolution) [13], a lightweight variant that removes the nonlinear activations and collapses the weight matrices into a single linear transformation, and (ii) **APPNP** (Approximate Personalized Propagation of Neural Predictions) [14], which decouples transformation from propagation, with a neural network producing node predictions that are then propagated via personalized PageRank, enabling a larger receptive field while reducing the risk of over-smoothing.

Both models share the same underlying propagation operator, which is built from the graph's adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, where n is the number of nodes and a nonzero entry A_{ij} encodes an edge between nodes i and j . Following the standard Graph Convolutional Network (GCN) derivation [15], self-loops are first added via $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, where $\mathbf{I}$ is the identity matrix, so that each node retains its own representation during aggregation, and the result is symmetrically normalized as $\hat{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}} \tilde{\mathbf{A}} \mathbf{D}^{-\frac{1}{2}}$, where $\mathbf{D}$ is the diagonal degree matrix of $\tilde{\mathbf{A}}$, with $D_{ii} = \sum_j \tilde{A}_{ij}$. Thus, while $\mathbf{A}$ fixes the topology over which information flows, it is the matrix $\mathbf{D}$ that weights how strongly each neighbor contributes to that flow.

Conventionally, these weights are derived purely from degree centrality. For an image i , the degree is simply the size of its neighborhood, $\delta(i) = |\mathcal{N}(i)|$. This makes normalization a strictly local and static count, reflecting how many neighbors a node has while remaining blind to how relevant those neighbors are in the feature space.

To overcome this limitation, the standard degree centrality in $\mathbf{D}$ is replaced by GRaNDe [11] in both SGC and APPNP. Each node (image) q receives a score that modulates its message amplification, with lower values providing stronger influence:

$$\delta_{\text{GRaNDe}}(q) = \delta_{\text{ctr}}(q) + \frac{1}{\delta_{\text{ctr}}(q)} \sum_{i \in \mathcal{N}(q)} \left[\exp\left(-\frac{\rho'(q, i)^2}{\sigma}\right) \right]^{-1},$$

where $\delta_{\text{ctr}}(q) = |\mathcal{N}(q)|$ is the degree centrality, which preserves the usual intuition of attenuating high-degree nodes. Division by zero never occurs here, as self-loops ensure $\delta_{\text{ctr}} \geq 1$ for every node.

The second term of the sum introduces a dynamic neighborhood penalty utilizing the inverse of a Gaussian Radial Basis Function (RBF). The variable $\rho'(q, i) \in [0, 1]$ denotes the min-max normalized Euclidean distance computed from the latent feature representations generated by the GNN's linear layer. Because these distances are measured in a learned latent space, the penalty is continuously recomputed each epoch during training. This allows GRaNDe to dynamically adapt to the evolving latent feature space: nodes with distant or dissimilar neighbors incur a larger penalty, which increases their overall effective degree and attenuates their message propagation. In contrast, nodes surrounded by close, highly relevant neighbors receive a lower penalty, enabling their learned representations to exert a stronger influence during message propagation.

Based on the final representations generated by the model, a softmax activation function is applied to produce class probabilities. The model is optimized in a semi-supervised setting using a small set of labeled nodes. The training process minimizes the standard classification error computed over these known examples, iteratively updating both the internal

weights of the network and the dynamic penalty terms of the normalization process.

III. EXPERIMENTAL EVALUATION

In this section, we present the experimental evaluation conducted on three publicly available datasets, as follows:

- **Flowers17** [16]: contains 1,360 images evenly distributed across 17 flower categories. Each class includes 80 images captured under diverse backgrounds and viewpoints, resulting in moderate intra-class variability.
- **Pets** [17]: contains 7,390 images across 37 pet classes with diverse poses, lighting, and backgrounds.
- **CUB200** [18]: includes 11,788 images of 200 bird species, providing detailed annotations commonly used for fine-grained recognition and localization tasks.

For each dataset, experiments were conducted using three feature extraction models pretrained on the ImageNet dataset:

- **ResNet152** [19]: a deep residual CNN. We used the 2048-dimensional feature vector after global average pooling of the final convolutional stage, before the classification layer.
- **SENet154** [20]: a ResNet variant with channel-wise attention. We used the 2048-dimensional feature vector after global average pooling of the final squeeze-and-excitation residual block, before the classification layer.
- **ViT-B16** [21]: a transformer model operating on patch sequences. We used the 768-dimensional class token from the last encoder layer.

The proposed framework was evaluated using three distinct GNN architectures: **SGC**, **APPNP**, and **Graph Attention Network v2 (GATv2)** [22]. SGC and APPNP employ degree-based propagation, allowing GRaNDe to be directly integrated into their normalization operators. GATv2 instead assigns learned attention coefficients to neighboring nodes, and only its underlying graph topology is varied in our experiments. It is included as a widely adopted attention-based GNN baseline to contextualize the effectiveness of D-GRaNDe under an alternative neighbor-weighting strategy.

Regarding the experimental setup, all GNN models were trained for 200 epochs. The learning rate was set to 10^{-3} across all datasets, except for CUB200, which utilized a learning rate of 10^{-2} . In terms of capacity, both SGC and APPNP were configured with 256 neurons in their hidden layer. For GATv2, this hidden-layer capacity was distributed across a multi-head attention mechanism consisting of 8 attention heads. The remaining network hyperparameters followed the default settings provided by the PyG library.

For the graph construction stage, we set $k = 40$ for both the depth of the kNN baseline graphs and the computation of the rank correlation measures. For DGCG, the edge selection scope was defined as $L = 200$ as done in the original publication. In this evaluation, we restricted the correlation measures to RBO and JaccardMax. Whenever the specific correlation graph is not explicitly stated in the results, RBO is considered the default topology. For the dynamic centrality configurations (GRaNDe and D-GRaNDe), the scaling parameter σ was evaluated within the interval $[0.1, 0.9]$. In experiments where σ is not specified, the reported values correspond to the best-performing σ configuration.

All experiments followed a 10-fold cross-validation protocol, in which each fold was used once as the training set and the remaining nine as the test set. Notably, this configuration trains the model on only 10% of the data while testing on the remaining 90%, creating a particularly challenging semi-supervised scenario with a small set of labels for training the

models. Each reported result represents the mean accuracy and the sample standard deviation computed across 5 independent executions.

Regarding efficiency, both G_{Ra}NDe and DGCG scale linearly with the dataset size for fixed L and k : the correlation graph is computed only once before training, and the dynamic normalization adds a cost per epoch proportional to the number of edges, as detailed in the Supplementary Material.

A. Hyperparameter Analysis

Two key hyperparameters of DGCG and G_{Ra}NDe are the graph density (η) and the scaling parameter for the Gaussian distance (σ), respectively. Since DGCG and G_{Ra}NDe are used together in D-G_{Ra}NDe, we jointly evaluate the impact of both hyperparameters on accuracy. Figure 2 presents one accuracy curve per σ value across densities. Notice that the highest accuracy is achieved for sparser graphs (low η). The curves are consistently ordered by σ , with $\sigma = 0.2$ being frequently the best across densities. Therefore, we adopted the density interval $\eta \in [0.03, 0.04]$ and $\sigma = 0.2$ as defaults, following the values in the literature. A relevant finding is that using DGCG and G_{Ra}NDe together does not seem to alter the individual behavior of these hyperparameters.

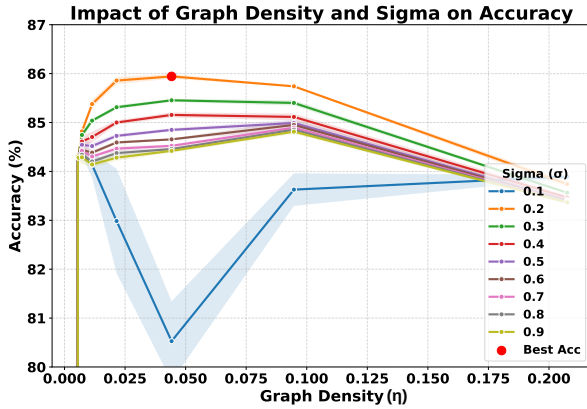


Fig. 2. Impact of the density coefficient (η) and σ on the accuracy of the SGC model using ResNet features on the Flowers dataset. Shaded bands denote the standard deviation and the red dot marks the best accuracy.

B. Ablation Study

To evaluate the individual contribution of the graph construction and degree normalization, we conducted an ablation study across different GNN architectures. Table I presents the accuracy obtained on the Flowers and CUB200 datasets using ResNet features, isolating the graph topology (Reciprocal vs. DGCG with RBO Correlation Graph) and the centrality measure used for normalization.

The direct comparison among traditional centralities (i.e., degree, closeness, and betweenness) reveals nearly identical accuracy across scenarios. When we replace these standard measures with G_{Ra}NDe over the Reciprocal (Rec.) graph, it provides substantial accuracy gains. For instance, using APPNP on the CUB200 dataset, the use of G_{Ra}NDe elevates the accuracy from 49.49% (standard degree) to 54.58%.

Notice that employing the DGCG graph with traditional degree normalization provides competitive results. However, the highest overall accuracy across both GNNs (SGC and APPNP) and datasets is achieved when D-G_{Ra}NDe (DGCG+G_{Ra}NDe) is employed. The best accuracy is observed with the APPNP architecture, reaching 87.80% on Flowers and 57.33% on

TABLE I
CLASSIFICATION ACCURACY (%) ON THE FLOWERS AND CUB200 DATASETS USING RESNET FEATURES, EVALUATED ACROSS GNN MODELS, GRAPH CONSTRUCTION METHODS, AND CENTRALITY MEASURES. **D-GRANDE IS GIVEN BY DGCG+GRANDE IN GRAY.**

GNN	Graph	Centrality Measure	Flowers	CUB200
GATV2	Rec	–	83.57 $\pm$ 0.14	43.92 $\pm$ 0.32
	DGCG	–	84.44 $\pm$ 0.22	40.57 $\pm$ 0.30
SGC	Rec	Degree	83.97 $\pm$ 0.04	53.80 $\pm$ 0.02
	Rec	Closeness	83.99 $\pm$ 0.04	53.79 $\pm$ 0.04
	Rec	Betweenness	83.98 $\pm$ 0.04	53.79 $\pm$ 0.03
	Rec	G _{Ra} NDe	85.11 $\pm$ 0.03	54.45 $\pm$ 0.03
	DGCG	Degree	84.29 $\pm$ 0.04	53.42 $\pm$ 0.04
	DGCG	Closeness	84.42 $\pm$ 0.09	53.43 $\pm$ 0.02
	DGCG	Betweenness	84.39 $\pm$ 0.08	53.43 $\pm$ 0.03
	DGCG	G_{Ra}NDe	85.97 $\pm$ 0.05	55.71 $\pm$ 0.01
APPNP	Rec	Degree	84.43 $\pm$ 0.09	49.49 $\pm$ 0.96
	Rec	Closeness	84.55 $\pm$ 0.11	49.51 $\pm$ 0.40
	Rec	Betweenness	84.47 $\pm$ 0.16	49.43 $\pm$ 0.31
	Rec	G _{Ra} NDe	86.19 $\pm$ 0.12	54.58 $\pm$ 0.24
	DGCG	Degree	84.21 $\pm$ 0.23	51.73 $\pm$ 0.09
	DGCG	Closeness	84.19 $\pm$ 0.11	51.73 $\pm$ 0.08
	DGCG	Betweenness	84.23 $\pm$ 0.63	51.73 $\pm$ 0.13
	DGCG	G_{Ra}NDe	87.80 $\pm$ 0.08	57.33 $\pm$ 0.09

CUB200. This corroborates our core hypothesis: integrating DGCG’s graph with G_{Ra}NDe’s dynamic weighting can lead to effectiveness improvements. Although both components are derived from rank information, they are complementary rather than redundant: the gain from G_{Ra}NDe depends on the topology it normalizes, indicating that graph construction and degree normalization should be considered together. Notice that GATv2, a widely adopted attention-based GNN, weights neighbors through learned attention instead of degree normalization, reaching 83.57% on Flowers and 43.92% on CUB200. As only its topology is varied, this configuration does not isolate the effect of degree normalization as in the SGC and APPNP settings, and is reported to contextualize D-G_{Ra}NDe under an alternative neighbor-weighting strategy.

C. Comparative Effectiveness of D-G_{Ra}NDe

Initially, we compare D-G_{Ra}NDe (DGCG+G_{Ra}NDe) against two traditional and commonly used graph constructions as baselines: k -nearest neighbors (kNN) and reciprocal kNN, both using the standard degree normalization. Table II presents the results.

Notice that across all datasets and feature extractors, D-G_{Ra}NDe consistently achieves higher accuracy than both baselines. It is important to highlight the high relative gains over the standard kNN graph on the CUB200 dataset. These gains are even more noticeable for the APPNP network: in some cases, the kNN graph fails almost entirely (e.g., 17.06% for ResNet and 29.97% for ViT features), with large standard deviations indicating unstable and inconsistent results. This happens because the standard kNN graph produces noisy and highly asymmetric edges, leading to poor connectivity that APPNP’s deeper propagation amplifies rather than smooths, causing the prediction quality to collapse. In contrast, D-G_{Ra}NDe does not suffer from these issues.

Table III compares D-G_{Ra}NDe, which combines DGCG and G_{Ra}NDe, against DGCG using the standard degree and G_{Ra}NDe using the reciprocal graph, as originally proposed in the literature. The objective is to assess whether the integration of these two approaches provides improvements in classification effectiveness across the evaluated datasets and features.

The results indicate that D-GRaNDe achieves the highest effectiveness in the majority of cases. On the CUB200 dataset, for instance, using ViT features and the APPNP model, D-GRaNDe achieves an accuracy of 78.09%, surpassing both DGCG (75.10%) and GRaNDe (74.41%). The behavior observed for the correlation graphs is also important: RBO consistently provides the highest accuracy under D-GRaNDe relative to DGCG, suggesting that RBO constitutes a reliable choice when a single correlation measure must be adopted. The only exception occurs on Pets with ViT features and APPNP, where DGCG remains slightly superior. Statistical tests and effect-size analyses for all comparisons are reported in the Supplementary Material.

We also conducted an experiment to see how D-GRaNDe compares with these approaches in different fold configurations (5, 10, 20, and 30 folds). Figure 3 presents a visualization with this purpose. In this context, more folds correspond to a considerably more challenging semi-supervised scenario, as fewer labeled instances are available to guide the learning process. As expected, the accuracy naturally decreases for all evaluated methods as the number of folds increases. However, the heatmaps illustrate that D-GRaNDe consistently sustains the highest accuracy levels across all configurations for both datasets. This consistent result across multiple splits reveals that D-GRaNDe mitigates the impact of limited labels.

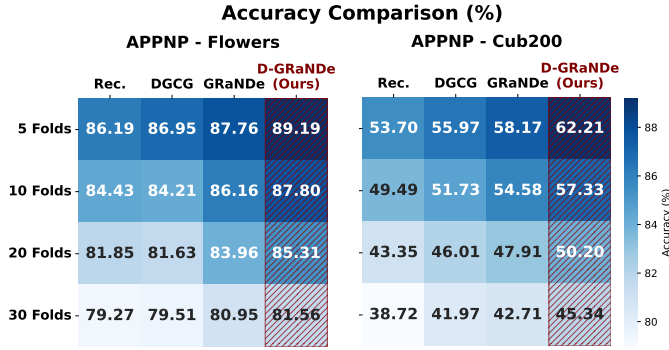


Fig. 3. Accuracy comparison (%) of the proposed D-GRaNDe method against baselines across different fold configurations for the APPNP model. Experiments were conducted on the Flowers and CUB200 datasets using ResNet features.

D. Comparison with Other Approaches

Finally, Table IV compares D-GRaNDe against traditional and state-of-the-art methods under the same experimental protocol (10% labeled data) for the semi-supervised image classification task. For D-GRaNDe, we report the highest accuracy achieved across the SGC and APPNP networks. The baselines encompass traditional classifiers (SVM, OPF [23], SL-Perc., and ML-Perc.), widely adopted semi-supervised strategies like label spreading (LS) and pseudolabeling (PL), and recent graph-based models (GATv2 [22], GNN-LDS [24], WSEF [25], ManifoldGCN [7], PolyNormer [26], DGCG [10], and GRaNDe [11]). Baselines without explicit citations refer to standard implementations from the *scikit-learn* library. Note that CoMatch [27] operates directly on input images instead of extracted features, thus yielding the same accuracy across columns. Overall, D-GRaNDe achieved the highest accuracy in five of the six dataset/feature combinations, being surpassed only by CoMatch on Flowers with SENet features. Notice that PolyNormer, a Graph Transformer among the baselines, remains below D-GRaNDe, with its hyperparameters reported in the Supplementary Material.

IV. CONCLUSION

In this work, we introduced D-GRaNDe, a unified framework for semi-supervised image classification that integrates the density-guided correlation graph (DGCG) with dynamic, rank-based degree normalization (GRaNDe). Unlike conventional GNNs, which build edges from raw feature distances and normalize propagation with a static degree count, D-GRaNDe incorporates rank information at both stages: edges are derived from the rank correlation between similarity rankings, and message passing is reweighted by the Gaussian-weighted distances to ranked neighbors. The results support our hypothesis: constructing a sparse, rank-correlation graph and dynamically normalizing its propagation through Gaussian-weighted neighbor rankings are highly complementary strategies that reinforce each other, leading to higher effectiveness.

As future work, we intend to extend the evaluation to larger datasets, including other modalities (e.g., audio and video) and additional GNN models, and to investigate adaptively selecting the correlation measure and the parameter σ during training.

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TABLE II
ACCURACY (%) COMPARING D-GRANDE WITH KNN AND KNN REC. GRAPHS ACROSS TWO GNNs. FOR EACH CORRELATION, WE REPORT THE ACCURACY USING THE DEFAULT AND THE BEST CORRESPONDING σ VALUE. THE BEST RESULT PER ROW IS IN **BOLD**, AND THE BEST PER FEATURE/DATASET COMBINATION IS HIGHLIGHTED IN **BLUE**. RELATIVE GAINS ARE REPORTED OVER KNN AND REC KNN GRAPHS.

Dataset	Feature	GNN	Accuracy with Degree Centrality		Accuracy with D-GRaNdE (Ours)									Relative Gains Over	
			kNN	kNN Rec	RBO			JaccardMax			Default $\sigma = 0.2$	Best σ	σ	kNN	Rec kNN
					Default $\sigma = 0.2$	Best σ	σ	Default $\sigma = 0.2$	Best σ	σ					
Flowers17	ResNet	SGC	79.56 $\pm$ 0.11	83.97 $\pm$ 0.04	85.97 $\pm$ 0.05	85.97 $\pm$ 0.05	0.2	85.61 $\pm$ 0.03	85.72 $\pm$ 0.73	0.1				+8.06%	+2.38%
		APPNP	78.10 $\pm$ 0.32	84.45 $\pm$ 0.24	87.80 $\pm$ 0.08	87.80 $\pm$ 0.08	0.2	85.96 $\pm$ 0.10	86.37 $\pm$ 0.12	0.1				+12.42%	+3.97%
	SENet	SGC	73.02 $\pm$ 0.10	76.15 $\pm$ 0.07	79.47 $\pm$ 0.09	79.47 $\pm$ 0.09	0.2	79.44 $\pm$ 0.04	79.68 $\pm$ 0.23	0.1				+9.12%	+4.64%
		APPNP	71.11 $\pm$ 0.22	77.90 $\pm$ 0.24	81.23 $\pm$ 0.11	81.47 $\pm$ 0.17	0.1	80.78 $\pm$ 0.10	80.93 $\pm$ 0.09	0.1				+14.57%	+4.58%
Pets	ResNet	SGC	92.81 $\pm$ 0.06	96.93 $\pm$ 0.02	98.57 $\pm$ 0.02	98.57 $\pm$ 0.02	0.2	98.06 $\pm$ 0.02	98.06 $\pm$ 0.23	0.1				+6.21%	+1.69%
		APPNP	91.67 $\pm$ 0.19	97.11 $\pm$ 0.02	96.10 $\pm$ 0.21	98.36 $\pm$ 0.02	0.4	91.56 $\pm$ 0.39	97.90 $\pm$ 0.05	0.4				+7.30%	+1.29%
	SENet	SGC	87.99 $\pm$ 0.02	89.86 $\pm$ 0.03	91.13 $\pm$ 0.01	91.13 $\pm$ 0.01	0.2	90.83 $\pm$ 0.03	90.83 $\pm$ 0.03	0.2				+3.57%	+1.41%
		APPNP	87.72 $\pm$ 0.05	90.09 $\pm$ 0.11	91.50 $\pm$ 0.05	91.50 $\pm$ 0.05	0.2	91.11 $\pm$ 0.06	91.11 $\pm$ 0.02	0.1				+4.31%	+1.57%
CUB200	ResNet	SGC	85.32 $\pm$ 0.03	89.68 $\pm$ 0.03	89.98 $\pm$ 0.03	90.41 $\pm$ 0.03	0.1	89.90 $\pm$ 0.01	90.24 $\pm$ 0.02	0.1				+5.97%	+0.81%
		APPNP	84.81 $\pm$ 0.13	89.73 $\pm$ 0.04	90.58 $\pm$ 0.04	90.58 $\pm$ 0.04	0.2	90.42 $\pm$ 0.04	90.53 $\pm$ 0.03	0.1				+6.80%	+0.95%
	SENet	SGC	84.88 $\pm$ 0.08	88.23 $\pm$ 0.04	90.26 $\pm$ 0.02	90.26 $\pm$ 0.02	0.2	90.09 $\pm$ 0.03	90.09 $\pm$ 0.03	0.2				+6.34%	+2.30%
		APPNP	80.87 $\pm$ 0.17	87.82 $\pm$ 0.06	85.56 $\pm$ 0.15	89.38 $\pm$ 0.06	0.4	87.48 $\pm$ 0.06	88.57 $\pm$ 0.05	0.5				+10.52%	+1.78%
CUB200	ResNet	SGC	47.04 $\pm$ 0.05	53.80 $\pm$ 0.02	54.53 $\pm$ 0.04	55.71 $\pm$ 0.01	0.3	56.48 $\pm$ 0.03	56.48 $\pm$ 0.03	0.2				+20.07%	+4.98%
		APPNP	17.06 $\pm$ 2.20	49.49 $\pm$ 0.48	57.33 $\pm$ 0.09	57.33 $\pm$ 0.09	0.2	56.46 $\pm$ 0.12	56.46 $\pm$ 0.12	0.2				+236.05%	+15.84%
	SENet	SGC	36.23 $\pm$ 0.06	40.31 $\pm$ 0.01	42.56 $\pm$ 0.01	42.56 $\pm$ 0.01	0.2	44.69 $\pm$ 0.01	44.69 $\pm$ 0.01	0.2				+23.35%	+10.87%
		APPNP	28.62 $\pm$ 0.56	39.02 $\pm$ 0.14	44.32 $\pm$ 0.10	44.46 $\pm$ 0.08	0.1	44.26 $\pm$ 0.14	44.63 $\pm$ 0.07	0.1				+55.94%	+14.38%
CUB200	ResNet	SGC	73.87 $\pm$ 0.05	78.36 $\pm$ 0.04	78.71 $\pm$ 0.01	81.94 $\pm$ 0.01	0.4	80.09 $\pm$ 0.01	80.09 $\pm$ 0.01	0.2				+10.92%	+4.57%
		APPNP	29.97 $\pm$ 4.07	71.42 $\pm$ 0.21	64.25 $\pm$ 0.36	78.09 $\pm$ 0.16	0.5	65.57 $\pm$ 6.01	73.14 $\pm$ 0.10	0.4				+160.56%	+9.34%

TABLE III
ACCURACY (%) COMPARING D-GRANDE WITH DGCG AND GRANDE ACROSS TWO GNNs. FOR EACH CORRELATION, WE REPORT THE ACCURACY USING THE DEFAULT AND THE BEST CORRESPONDING σ VALUE. THE BEST RESULT PER ROW IS IN **BOLD**, AND THE BEST PER FEATURE/DATASET COMBINATION IS HIGHLIGHTED IN **BLUE**. RELATIVE GAINS ARE REPORTED OVER DGCG AND GRANDE.

Dataset	Feature	GNN	Accuracy with DGCG		Accuracy with GRaNdE		Accuracy with D-GRaNdE (Ours)									Relative Gains Over	
			RBO	JaccardMax	Best σ	σ	RBO			JaccardMax			Default $\sigma = 0.2$	Best σ	σ	DGCG	GRaNdE
							Default $\sigma = 0.2$	Best σ	σ	Default $\sigma = 0.2$	Best σ	σ					
Flowers17	ResNet	SGC	84.24 $\pm$ 0.06	85.58 $\pm$ 0.12	85.11 $\pm$ 0.021	0.2	85.97 $\pm$ 0.05	85.97 $\pm$ 0.05	0.2	85.61 $\pm$ 0.03	85.72 $\pm$ 0.73	0.1				+0.46%	+1.01%
		APPNP	84.24 $\pm$ 0.16	85.30 $\pm$ 0.19	86.19 $\pm$ 0.121	0.2	87.80 $\pm$ 0.08	87.80 $\pm$ 0.08	0.2	85.96 $\pm$ 0.10	86.37 $\pm$ 0.12	0.1				+2.93%	+1.87%
	SENet	SGC	78.40 $\pm$ 0.10	77.28 $\pm$ 0.13	77.32 $\pm$ 0.048	0.2	79.47 $\pm$ 0.09	79.47 $\pm$ 0.09	0.2	79.44 $\pm$ 0.04	79.68 $\pm$ 0.23	0.1				+1.63%	+3.05%
		APPNP	78.26 $\pm$ 0.18	77.98 $\pm$ 0.14	79.91 $\pm$ 0.097	0.2	81.23 $\pm$ 0.11	81.47 $\pm$ 0.17	0.1	80.78 $\pm$ 0.10	80.93 $\pm$ 0.09	0.1				+4.10%	+1.95%
Pets	ResNet	SGC	98.01 $\pm$ 0.03	98.05 $\pm$ 0.05	97.76 $\pm$ 0.014	0.1	98.57 $\pm$ 0.02	98.57 $\pm$ 0.02	0.2	98.06 $\pm$ 0.02	98.06 $\pm$ 0.23	0.1				+0.53%	+0.83%
		APPNP	97.84 $\pm$ 0.08	97.83 $\pm$ 0.12	98.18 $\pm$ 0.048	0.2	96.10 $\pm$ 0.21	98.36 $\pm$ 0.02	0.4	91.56 $\pm$ 0.39	97.90 $\pm$ 0.05	0.4				+0.53%	+0.18%
	SENet	SGC	90.33 $\pm$ 0.04	90.18 $\pm$ 0.01	90.79 $\pm$ 0.004	0.1	91.13 $\pm$ 0.01	91.13 $\pm$ 0.01	0.2	90.83 $\pm$ 0.03	90.83 $\pm$ 0.03	0.2				+0.89%	+0.37%
		APPNP	90.18 $\pm$ 0.10	90.26 $\pm$ 0.04	91.20 $\pm$ 0.042	0.2	91.50 $\pm$ 0.05	91.50 $\pm$ 0.05	0.2	91.11 $\pm$ 0.06	91.11 $\pm$ 0.02	0.1				+1.37%	+0.33%
CUB200	ResNet	SGC	89.68 $\pm$ 0.00	89.81 $\pm$ 0.02	89.98 $\pm$ 0.017	0.1	89.98 $\pm$ 0.03	90.41 $\pm$ 0.03	0.1	89.90 $\pm$ 0.01	90.24 $\pm$ 0.02	0.1				+0.67%	+0.48%
		APPNP	89.49 $\pm$ 0.06	89.98 $\pm$ 0.07	90.56 $\pm$ 0.037	0.2	90.58 $\pm$ 0.04	90.58 $\pm$ 0.04	0.2	90.42 $\pm$ 0.04	90.53 $\pm$ 0.03	0.1				+0.67%	+0.02%
	SENet	SGC	89.51 $\pm$ 0.01	89.76 $\pm$ 0.02	88.34 $\pm$ 0.023	0.2	90.26 $\pm$ 0.02	90.26 $\pm$ 0.02	0.2	90.09 $\pm$ 0.03	90.09 $\pm$ 0.03	0.2				+0.56%	+2.17%
		APPNP	89.36 $\pm$ 0.07	89.56 $\pm$ 0.07	88.50 $\pm$ 0.039	0.3	85.56 $\pm$ 0.15	89.38 $\pm$ 0.06	0.4	87.48 $\pm$ 0.06	88.57 $\pm$ 0.05	0.5				-0.20%	+0.99%
CUB200	ResNet	SGC	53.40 $\pm$ 0.03	53.73 $\pm$ 0.03	54.45 $\pm$ 0.027	0.4	54.53 $\pm$ 0.04	55.71 $\pm$ 0.01	0.3	56.48 $\pm$ 0.03	56.48 $\pm$ 0.03	0.2				+5.12%	+3.73%
		APPNP	51.73 $\pm$ 0.12	50.43 $\pm$ 0.16	54.58 $\pm$ 0.238	0.2	57.33 $\pm$ 0.09	57.33 $\pm$ 0.09	0.2	56.46 $\pm$ 0.12	56.46 $\pm$ 0.12	0.2				+10.83%	+5.04%
	SENet	SGC	39.72 $\pm$ 0.03	40.76 $\pm$ 0.03	41.79 $\pm$ 0.013	0.3	42.56 $\pm$ 0.01	42.56 $\pm$ 0.01	0.2	44.69 $\pm$ 0.01	44.69 $\pm$ 0.01	0.2				+9.64%	+6.94%
		APPNP	38.65 $\pm$ 0.13	39.94 $\pm$ 0.10	43.78 $\pm$ 0.178	0.2	44.32 $\pm$ 0.10	44.46 $\pm$ 0.08	0.1	44.26 $\pm$ 0.14	44.63 $\pm$ 0.07	0.1				+11.74%	+1.94%
CUB200	ResNet	SGC	81.25 $\pm$ 0.02	80.77 $\pm$ 0.02	78.58 $\pm$ 0.013	0.4	78.71 $\pm$ 0.01	81.94 $\pm$ 0.01	0.4	80.09 $\pm$ 0.01	80.09 $\pm$ 0.01	0.2				+0.85%	+4.28%
		APPNP	75.10 $\pm$ 0.25	64.58 $\pm$ 0.52	74.41 $\pm$ 0.222	0.4	64.25 $\pm$ 0.36	78.09 $\pm$ 0.16	0.5	65.57 $\pm$ 6.01	73.14 $\pm$ 0.10	0.4				+3.98%	+4.95%

TABLE IV
STATE-OF-THE-ART (SOTA) COMPARISON (ACCURACY %) USING THE SAME FEATURES AND 10% OF TRAINING DATA. BEST RESULTS FOR EACH DATASET AND FEATURE EXTRACTOR ARE SHOWN IN **BOLD BLUE**.

Method	Flowers			CUB200		
	ResNet	SENet	VIT-B16	ResNet	SENet	VIT-B16
kNN	63.67	48.71	91.91	36.67	22.23	56.62
LS+kNN	73.49	58.05	95.74	36.99	20.00	66.15
SVM	80.54	73.30	96.75	48.84	35.32	75.61
LS+SVM	73.53	59.84	94.49	38.70	24.82	66.81
SL-Perc.	75.44	71.84	75.79	39.91	36.39	70.84
LS+SL-Perc.	72.34	59.27	93.71	39.21	25.41	65.45
ML-Perc.	78.88	72.62	92.59	32.24	32.15	12.02
LS+ML-Perc.	73.03	59.39	95.13	39.68	25.72	62.81
PL+SGD	82.69	76.87	96.84	21.67	20.96	30.19
OPF [23] (2014)	71.77	64.00	96.50	38.59	30.94	73.27
LS+OPF [23] (2014)	72.66	59.25	94.22	39.28	25.38	66.68
CoMatch [27] (2021)	82.55	82.55	82.55	38.29	38.29	38.29
GNN-LDS [24] (2019)	79.32	73.69	96.66	37.78	—	52.42
WSEF [25] (2021)	85.12	76.16	97.82	52.17	36.49	78.64
GATv2 [22] (2022)	83.57	76.24	96.92	43.92	34.14	68.01
MGCN [7] (2023)	85.88	78.82	97.43	52.85	40.31	79.27
PolyNormer [26] (2024)	85.81	79.46	97.60	54.10	44.01	75.54
DGCG [10] (2025)	85.68	78.42	98.09	54.43	41.00	81.23
GRaNdE [11] (2026)	86.19	80.10	98.18	54.58	43.78	78.58
D-GRaNdE (Ours)	87.80	81.47	98.57	57.33	44.69	81.94

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